Precision, Recall, ROC and Cost-Based Decisions

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia (15 September 2026) Source: `Lectures/L10_student_precision_recall_roc_auc.pdf` (50 slides) · Previous: Metrics, thresholds and costs · Next: Support Vector Machines

> **How to use this file.** Lecture 9 introduced the metrics on 400 roads. Lecture 10 rebuilds them **slowly, one observation at a time**, on **8 roads** you can count on your fingers, and adds three new ideas: **precision depends on prevalence**, **why F1 uses the harmonic mean** (with a proof), and **expected cost per observation as prevalence changes** (with iso-cost lines that depend on π). Part A collects the maths; Part B walks all 50 slides; Part C is practice. If something here feels fast, the matching section of L09 Part A has the long version.

The one dataset for the whole lecture (sorted by score):

obs	A	B	C	D	E	F	G	H
true y	1	0	1	1	0	1	0	0
score s	.95	.85	.75	.65	.55	.45	.25	.15

Positives: A, C, D, F. Negatives: B, E, G, H. Note **B**: a negative with a high score.

What this lecture is really about:

$$\underbrace{\text{count errors}}_{\text{confusion matrix}} \rightarrow \underbrace{\text{assess ranking}}_{\text{ROC, AUC}} \rightarrow \underbrace{\text{choose actions using consequences}}_{\tau^*=C_{FP}/(C_{FP}+C_{FN}),\ R=C_{FP}(1-\pi)\text{FPR}+C_{FN}\pi(1-\text{TPR})}$$

Step-up path

You need. The confusion matrix and the idea of a threshold.

1. Compute metrics at one threshold, by counting.
2. Walk down the ranked list to draw the ROC curve by hand.
3. Read AUC as a probability about random pairs.
4. See why precision depends on prevalence, and ROC does not.
5. Connect a cost per case to a total cost per population.

You can stop when you can say which curve to report for a rare event, and against what baseline.

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Part A — Maths from the ground up

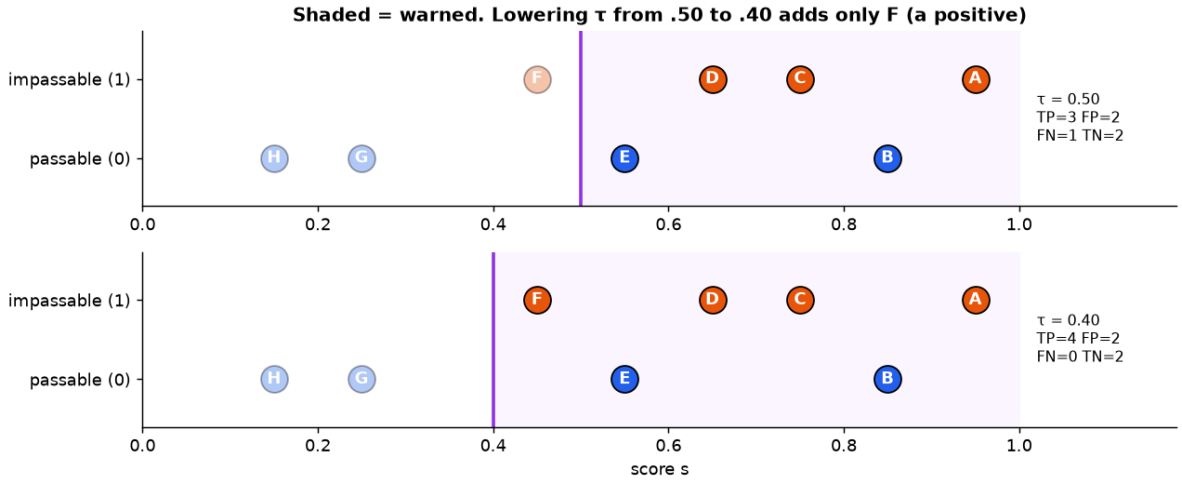
1. Threshold → confusion matrix, by counting
2. Recall vs precision: same numerator, different denominator
3. Precision depends on prevalence (Bayes' rule)
4. Arithmetic vs harmonic mean, and F1
5. ROC by walking the sorted list; AUC with ties
6. Expected cost of an action and the threshold
7. Cost per observation in a population
8. Class weights vs thresholds

Part B — Slides · All 50 slides

Part C — Practice · Jump

Part A Maths From the Ground Up

A1 Threshold → confusion matrix, by counting



The 8 roads on a score line. Shaded = warned. τ = 0.50 warns A–E; τ = 0.40 also warns F, a positive, so TP goes 3 → 4 and nothing else changes.

Decision rule: $\hat{y}_i = 1$ (warn) if $s_i \geq \tau$, else 0.

> **Recipe:** list who is warned → split them into actual 1 (TP) and actual 0 (FP) → the unwarned split into actual 1 (FN) and actual 0 (TN).

Full table for every threshold (lower τ just past each score):

τ	warned	TP	FP	FN	TN	accuracy	recall = TPR	FPR	precision	cost $2FP + 6FN$
> .95	—	0	0	4	4	4/8	0	0	undefined	24
.95	A	1	0	3	4	5/8	1/4	0	1	18
.85	A B	1	1	3	3	4/8	1/4	1/4	1/2	20
.75	A–C	2	1	2	3	5/8	2/4	1/4	2/3	14
.65	A–D	3	1	1	3	6/8	3/4	1/4	3/4	8
.55	A–E	3	2	1	2	5/8	3/4	2/4	3/5	10
.45	A–F	4	2	0	2	6/8	1	2/4	2/3	**4**
.25	A–G	4	3	0	1	5/8	1	3/4	4/7	6
.15	all	4	4	0	0	4/8	1	1	1/2	8

($\tau = 0.50$ gives the same decisions as 0.55; $\tau = 0.40$ the same as 0.45; $\tau = 0.60$ the same as 0.65.)

Monotonicity. Lowering τ can only **add** warnings, so TP and FP can only go up, FN and TN only down (slide 7). Recall never decreases. Precision can go either way.

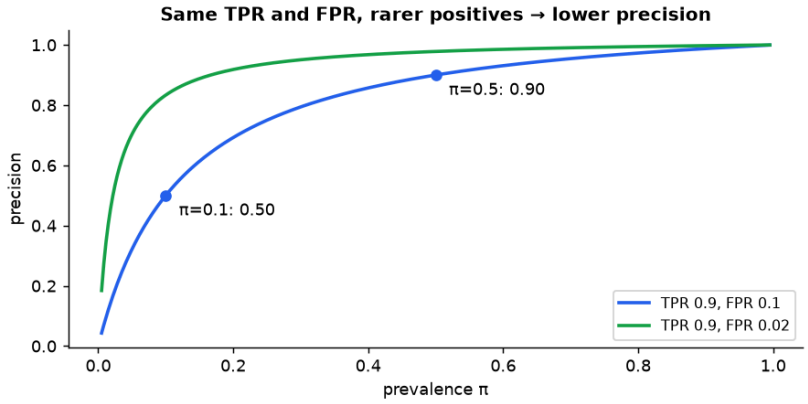
A2 Recall vs precision: same numerator, different denominator

	Recall	Precision
start from	**actual** positives: A, C, D, F	**predicted** positives (warnings): A, B, C, D, E
count successes	A, C, D (warned and positive)	A, C, D (warned and positive)
at τ = .50	$3/4 = 75\%$	$3/5 = 60\%$
asks	did we find the impassable roads?	were our warnings usually correct?
hurt by	a **missed** event (FN)	a **false** warning (FP)

What moves them as τ drops (from the A1 table):

- cross a **negative** (B at .85): recall unchanged, precision falls ($1 \to 1/2$);
- cross a **positive** (C, D at .75, .65): recall rises **and precision can rise** ($1/2 \to 3/4$).

A3 Precision depends on prevalence (Bayes' rule)



Precision vs prevalence for TPR 0.9 and FPR 0.1 (blue) or 0.02 (green): at π = 0.5 precision is 0.90; at π = 0.1 it falls to 0.50.

Let $\pi = \Pr(Y = 1)$. In a population of N:

- expected TP $= N\pi \cdot \mathrm{TPR}$ (positives × fraction caught);
- expected FP $= N(1-\pi) \cdot \mathrm{FPR}$ (negatives × fraction falsely warned).

$$\text{precision} = \frac{TP}{TP + FP} = \frac{\pi\,\mathrm{TPR}}{\pi\,\mathrm{TPR} + (1-\pi)\,\mathrm{FPR}}.$$

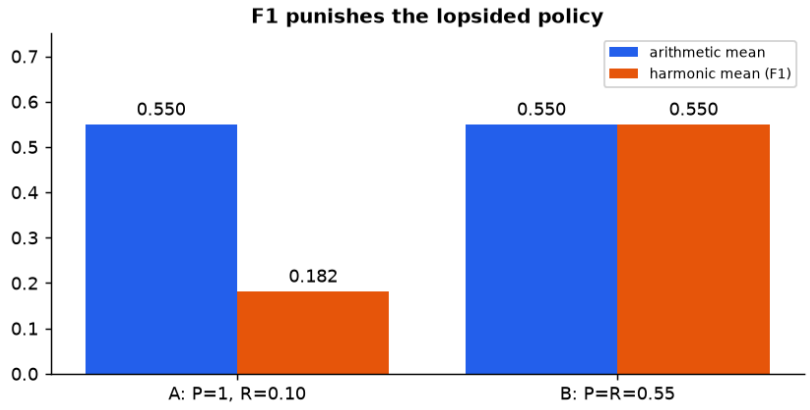
This is exactly **Bayes' rule** (L1, L3): $\Pr(Y = 1 \mid \text{warn}) = \frac{\Pr(\text{warn}|Y=1)\,\Pr(Y=1)}{\Pr(\text{warn})}$.

π	numerator 0.9π	$+\,0.1(1-\pi)$	precision
0.50	0.45	0.05	$0.45/0.50 = 0.90$
0.10	0.09	0.09	$0.09/0.18 = 0.50$
0.01	0.009	0.099	$0.009/0.108 = 0.083$

Same detector (same TPR, FPR), rarer positives ⇒ more of the warnings are false. TPR and FPR don't change with π because they divide by actual-class counts; precision does.

Baseline. If you warn regardless of the label (e.g. warn everyone, or warn at random), precision $= \pi$, not 0.5.

A4 Arithmetic vs harmonic mean, and F1



Policy A (P = 1, R = 0.10) and policy B (P = R = 0.55) have the same arithmetic mean 0.55; the harmonic mean scores A at 0.18.

Harmonic mean of $P, R > 0$: the reciprocal of the average reciprocal:

$$\mathrm{HM} = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P+R}.$$

AM ≥ HM, with the gap measured exactly.

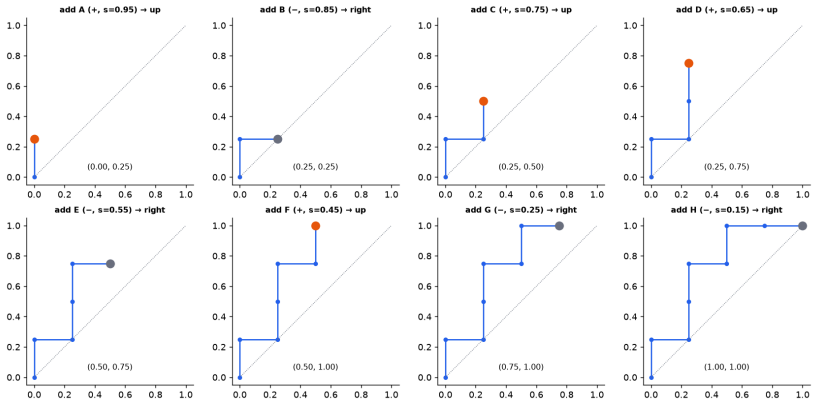
$$\mathrm{AM} - \mathrm{HM} = \frac{P+R}{2} - \frac{2PR}{P+R} = \frac{(P+R)^2 - 4PR}{2(P+R)} = \frac{P^2 - 2PR + R^2}{2(P+R)} = \frac{(P-R)^2}{2(P+R)} \geq 0.$$

Zero iff $P = R$. The more lopsided the pair, the bigger the penalty.

> **Example.** $P = 1$, $R = 0.10$: AM $= 0.55$, HM $= \frac{0.20}{1.10} = 0.182$. Gap $= \frac{0.81}{2.2} = 0.368$ ✓ $(0.55 - 0.182)$.

F1 in counts: $F_1 = \frac{2TP}{2TP+FP+FN}$. At $\tau = .50$: $\frac{6}{6+2+1} = \frac{2}{3}$. F1 **ignores TN** and **weighs FP and FN equally**. It's a design choice; when error costs are known, use them (A6).

A5 ROC by walking the sorted list; AUC with ties



ROC built one observation at a time: a positive moves up ¼, a negative moves right ¼.

$$\mathrm{TPR} = \frac{TP}{TP+FN}\ (\text{of actual positives}), \qquad \mathrm{FPR} = \frac{FP}{FP+TN}\ (\text{of actual negatives}), \qquad \text{specificit}$$

Walk the sorted list A→H: A(+) up, B(−) right, C(+) up, D(+) up, E(−) right, F(+) up, G(−) right, H(−) right. Points: $(0,0), (0,\frac{1}{4}), (\frac{1}{4},\frac{1}{4}), (\frac{1}{4},\frac{1}{2}), (\frac{1}{4},\frac{3}{4}), (\frac{1}{2},\frac{3}{4}), (\frac{1}{2},1), (\frac{3}{4},1), (1,1)$.

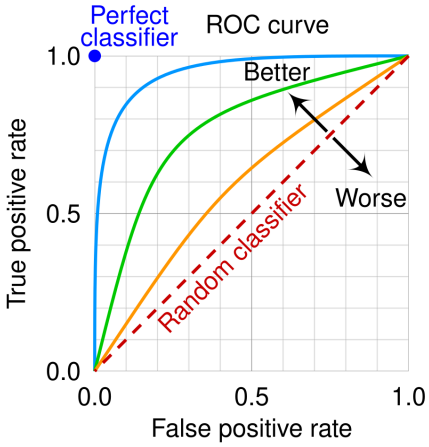
AUC two ways.

- **Area:** four columns of width ¼ with heights ¼, ¾, 1, 1 ⇒ $\frac{1}{4}(\frac{1}{4}+\frac{3}{4}+1+1) = \frac{12}{16} = 0.75$. (Vertical segments add no area.)
- **Pairs:** 4 × 4 = 16 (positive, negative) pairs; positive scored higher in 12 ⇒ $12/16 = 0.75$.

With ties:

$$\mathrm{AUC} = \Pr(s_+ > s_-) + \tfrac{1}{2}\Pr(s_+ = s_-) = \frac{\text{wins} + \frac{1}{2}\text{ties}}{n_+ n_-}.$$

(A tie makes the staircase take a diagonal step; the triangle under it is half a cell.) **AUC measures ranking. It is not the fraction of correct classifications.**



A generic ROC picture: the perfect classifier sits at (0, 1), random guessing on the diagonal, better curves bow toward the top-left. (Wikimedia Commons: Roc_curve.svg, CC BY-SA 4.0)

A6 Expected cost of an action and the threshold

Cost table (agreed decision units; illustrative, not measured safety values):

action	actually impassable	actually passable
issue warning	0	$C_{FP} = 2$
don't warn	$C_{FN} = 6$	0

For a road with calibrated $p = \Pr(Y = 1 \mid x)$:

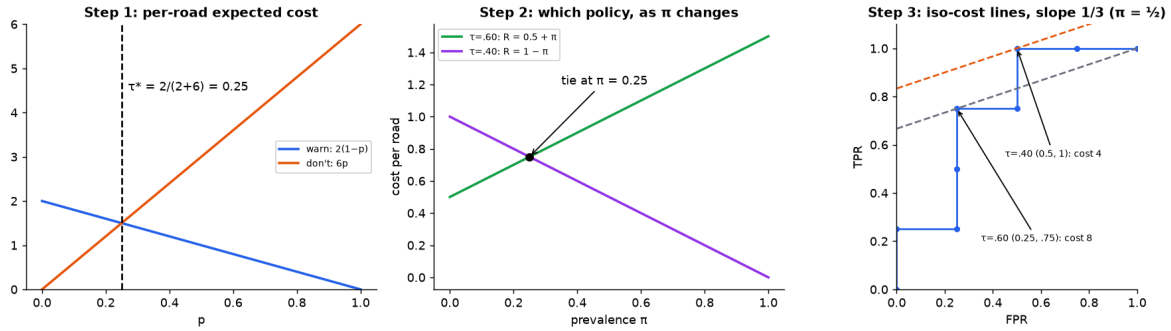
$$R(1 \mid x) = 0 \cdot p + C_{FP}(1 - p) = C_{FP}(1 - p), \qquad R(0 \mid x) = C_{FN}\, p + 0 \cdot (1 - p) = C_{FN}\, p.$$

Warn iff $R(1) \le R(0)$:

$$C_{FP}(1 - p) \le C_{FN} p \iff C_{FP} \le (C_{FP} + C_{FN}) p \iff p \ge \tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{2}{8} = \frac{1}{4}.$$

Equal costs ⇒ 0.5. **A bigger missed-event cost lowers the threshold.** (At equality either action is fine; the rule picks "warn".)

A7 Cost per observation in a population



Step 1: per-road costs cross at p = 0.25. Step 2: policies τ = .60 and τ = .40 as prevalence changes: they tie at π = 0.25. Step 3: iso-cost lines of slope 1/3 on the 8-road ROC.

Population of N with prevalence π. Assume the **class-conditional** TPR and FPR carry over.

quantity	expected value
actual negatives	$N(1-\pi)$
false positives	$N(1-\pi)\,\mathrm{FPR}$
actual positives	$N\pi$
false negatives	$N\pi(1-\mathrm{TPR})$

Multiply by costs, add, divide by N:

$$\boxed{R = C_{FP}(1-\pi)\,\mathrm{FPR} + C_{FN}\,\pi\,(1-\mathrm{TPR})} \quad \text{(expected cost per observation)}.$$

Same TPR and FPR, different π ⇒ different cost. That's why a policy validated in one district may be wrong in another.

Iso-cost lines. Fix $R = r$, write $u = \mathrm{FPR}$, $v = \mathrm{TPR}$, solve for v:

$$v = 1 - \frac{r}{C_{FN}\pi} + \frac{C_{FP}(1-\pi)}{C_{FN}\pi}\,u.$$

Slope $= \dfrac{C_{FP}(1-\pi)}{C_{FN}\pi}$. With 2, 6 and $\pi = \frac{1}{2}$: slope $\frac{1}{3}$. (L9's slope $\frac{c_{FP}n_-}{c_{FN}n_+}$ is the same thing with $n_-/n_+ = (1-\pi)/\pi$.) For a fixed FPR, higher TPR = lower cost. For an empirical (staircase) ROC, just compare the

achievable points; no smooth tangency needed.

A8 Class weights vs thresholds

Two different knobs that people often confuse:

	Weighted training	Threshold selection
what changes	the **objective** used to fit θ: $-\sum_i [w_1 y_i \log p_i + w_0(1-y_i)\log(1-p_i)]$	the **action rule** for an already-fitted model
when	before/during fitting	after fitting
outputs	p no longer estimates $\Pr(Y=1 \mid x)$ for the deployment population (it's tilted toward the up-weighted class)	p unchanged; only decisions move

Careful. Don't assume weighting and lowering the threshold are interchangeable, especially in a fitted, **regularised** model. If you use cost-based τ^*, it needs calibrated p, which weighting breaks.

Part B Slide by Slide (all 50 slides)

Slide 1 Title

Precision, Recall, ROC and Cost-Based Decisions. Prof. Udit Bhatia, 15 September 2026.

Slide 2 Where this fits in our course

1. Likelihood and logistic regression: estimate parameters (L6).
2. Regularisation: control the coefficients (L7–L8).
3. Confusion matrix and accuracy: count consequences (L9).
4. **Today:** vary the threshold, assess ranking, include error costs.

> **Summary:** the fitted model stays **fixed** throughout the worked example.

Slide 3 The decision problem

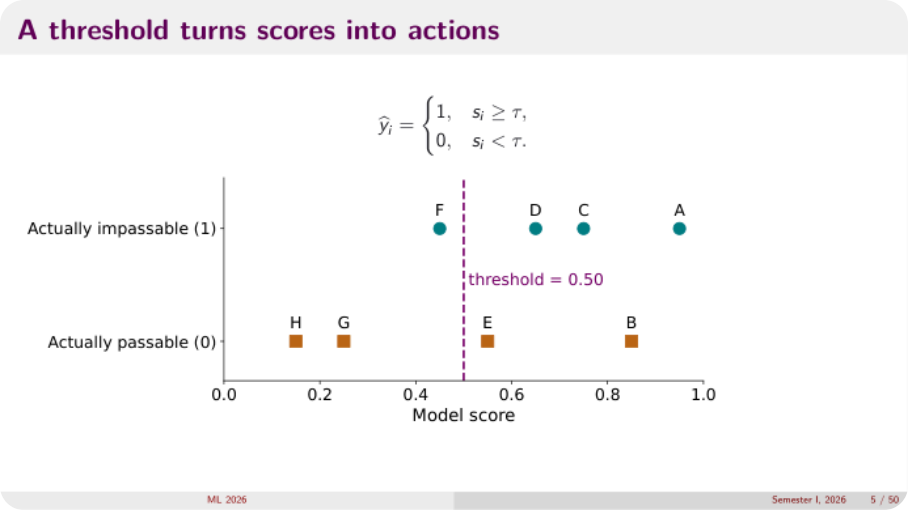
A road team decides whether to issue an impassability warning. $y = 1$: actually impassable; $y = 0$: passable. s: model score (higher = stronger evidence for 1). $\hat{y} = 1$: warn.

Eight held-out observations are a **teaching** dataset, not deployment evidence. **A score can be used for ranking without being a reliable probability.**

Slide 4 One dataset for the whole lecture

Table at the top of this note. Four positives (A, C, D, F), four negatives (B, E, G, H), already sorted by decreasing score. **Notice B**: a negative can score high (0.85).

Slide 5 A threshold turns scores into actions

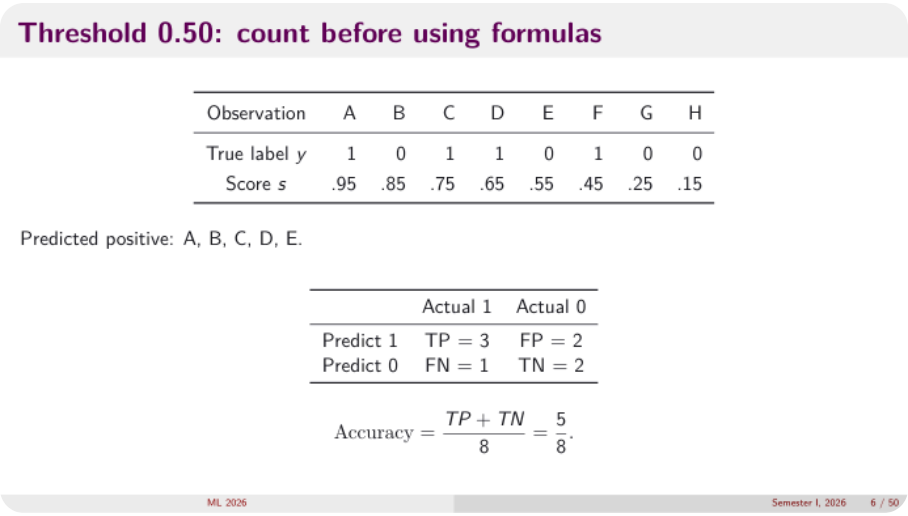


Slide 5 — Scores on a line, split by true class, with τ = 0.50.

$$\hat{y}_i = \begin{cases} 1, & s_i \geq \tau \\ 0, & s_i < \tau \end{cases}$$

At $\tau = 0.50$, everything right of the line is warned: A, B, C, D, E.

Slide 6 Threshold 0.50: count before using formulas



Slide 6 — The confusion matrix at τ = 0.50.

Predicted positive: A, B, C, D, E.

	actual 1	actual 0
predict 1	TP = 3 (A, C, D)	FP = 2 (B, E)
predict 0	FN = 1 (F)	TN = 2 (G, H)

Accuracy $= (TP + TN)/8 = 5/8$.

Slide 7

Threshold 0.40: what changed?

F (score 0.45) now gets a warning; no other observation changes.

	actual 1	actual 0
predict 1	TP = 4	FP = 2
predict 0	FN = 0	TN = 2

Insight: lowering the threshold **cannot decrease** TP or FP on a fixed dataset (A1).

Slide 8

Recall step 1: start with actual positives

Keep $\tau = .50$. Look **only** at the four actually impassable roads A, C, D, F. A, C, D were warned; F was missed. Of these four, how many did we catch? → 3.

Slide 9

Recall step 2: turn that count into a rate

Actual positives $= \underbrace{A, C, D}_{TP=3} + \underbrace{F}_{FN=1}$; $TP + FN = 4$.

$$\text{Recall} = \frac{\text{positives caught}}{\text{all actual positives}} = \frac{TP}{TP + FN} = \frac{3}{4} = 75\%.$$

Recall asks how much of the real problem we detected.

Slide 10 Precision step 1: start with issued warnings

Still $\tau = .50$. Look **only** at the five warned roads A, B, C, D, E. A, C, D correct; B, E false warnings. Of the five warnings, how many were correct? → 3.

Slide 11 Precision step 2: turn that count into a rate

Warnings $= \underbrace{A, C, D}_{TP=3} + \underbrace{B, E}_{FP=2}$; $TP + FP = 5$.

$$\text{Precision} = \frac{\text{correct warnings}}{\text{all warnings}} = \frac{TP}{TP + FP} = \frac{3}{5} = 60\%.$$

Precision asks how trustworthy the issued warnings were.

Slide 12 The numerator is the same; the denominator changes

Comparison table in A2. **A missed event affects recall. A false warning affects precision.**

Slide 13 Threshold step 1: a false warning lowers precision

τ	warned	recall	precision
.95	A	1/4	1/1 = 1
.85	A, B	1/4	1/2

B is negative: we added no detected positive, only a false warning. **Recall fixed, precision falls.**

Slide 14 Threshold step 2: correct warnings raise precision

τ	warned	recall	precision
.85	A, B	1/4	1/2
.65	A, B, C, D	3/4	3/4

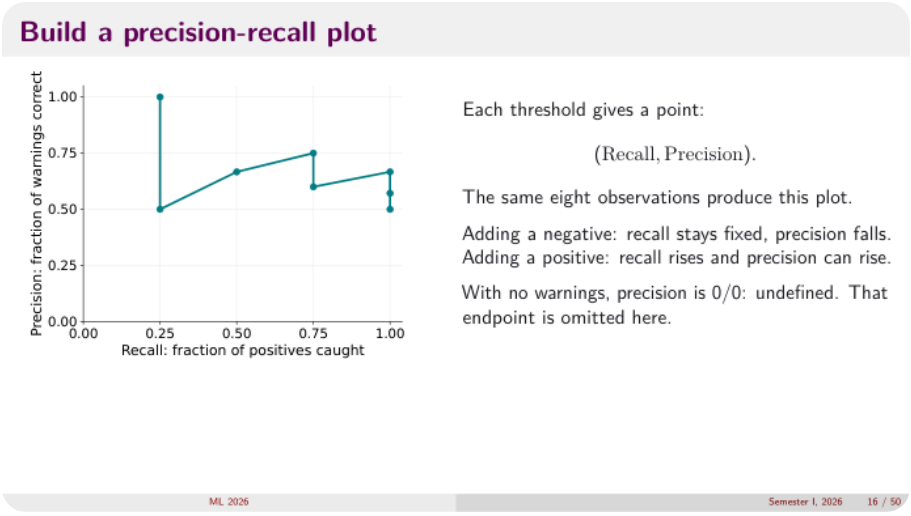
C and D are positives: **both** recall and precision rose. **Lowering a threshold does not force precision down at every step.**

Slide 15 Threshold step 3: recover the last missed positive

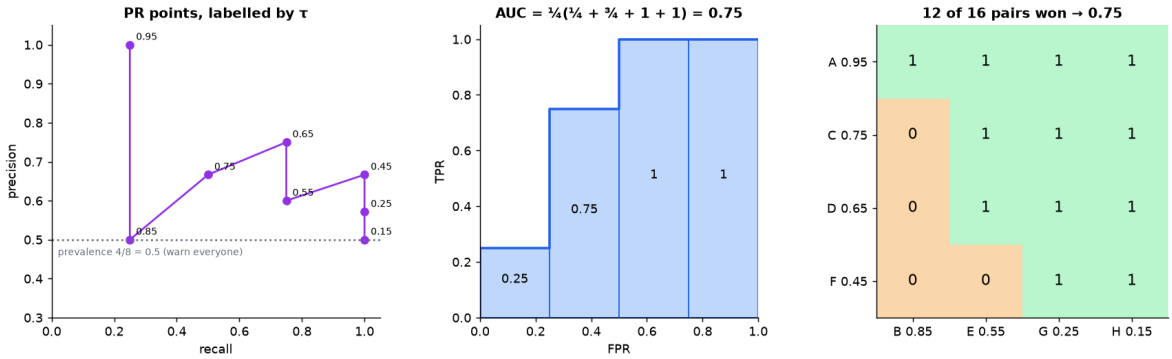
τ	TP	FP	recall	precision
.50	3	2	3/4	3/5
.40	4	2	4/4 = 1	4/6 = 2/3

Recall is now 100%, but B and E are still false warnings. Warn all eight: recall stays 1, precision falls to $4/8$ (the prevalence).

Slide 16 Build a precision-recall plot



Slide 16 — Each threshold gives one (recall, precision) point.



The same PR points, each labelled by its threshold, with the prevalence line; plus the AUC strips and the 4×4 pair table.

Points (recall, precision) by τ: .95 → (¼, 1); .85 → (¼, ½); .75 → (½, ⅔); .65 → (¾, ¾); .55 → (¾, ⅗); .45 → (1, ⅔); .25 → (1, 4/7); .15 → (1, ½).

- Adding a negative: recall fixed, precision falls (straight down).
- Adding a positive: recall rises and precision can rise (up-right).
- No warnings: precision 0/0, undefined; that endpoint is omitted.

Slide 17 Does a PR curve tell us how good a model is?

Two models on the same population, both catch 80 of 100 positives (recall 0.80).

	Model A	Model B
precision	0.80	0.50
total warnings = TP/precision	$80/0.8 = 100$	$80/0.5 = 160$
false warnings	20	80

At the same recall, higher precision means fewer false warnings for the same number of positives found.

Slide 18 Why precision changes with prevalence

$$\text{Precision} = \frac{\pi\,\text{TPR}}{\pi\,\text{TPR} + (1-\pi)\,\text{FPR}}.$$

TPR = .90, FPR = .10: $\pi = .50 \to .45/(.45 + .05) = .90$; $\pi = .10 \to .09/(.09 + .09) = .50$ (A3).

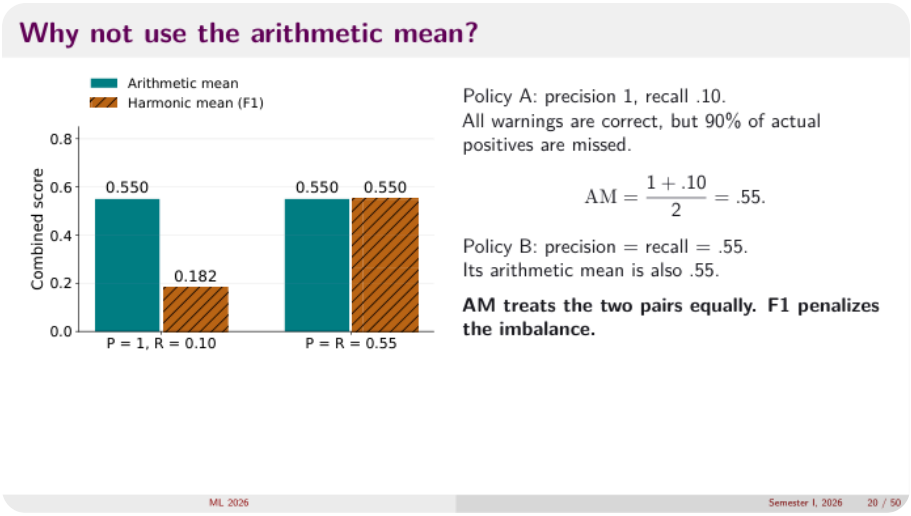
Fewer actual positives can mean a larger fraction of warnings are false, even when TPR and FPR are unchanged.

Slide 19 Reading a PR curve responsibly

- Higher precision at a given recall is better; higher recall at a given precision is better.
- If curves cross, the better model depends on the operating region and consequences.
- For selection independent of the label, population precision = prevalence π, **not always .50**.
- A high PR curve doesn't establish calibrated probabilities or choose a cost-optimal threshold.

Warning everyone gives recall 1 and precision π. Finite-sample random rankings fluctuate around that reference.

Slide 20 Why not use the arithmetic mean?



Slide 20 — Policy A (P = 1, R = .10) and policy B (P = R = .55): AM 0.55 for both, HM 0.182 vs 0.55.

Policy A: precision 1, recall .10: all warnings correct, but 90% of positives missed. AM $= (1 + .10)/2 = .55$. Policy B: P = R = .55, AM .55. **AM treats them equally; F1 penalises the imbalance.**

Slide 21 Why the harmonic mean penalizes imbalance

$\text{HM} = \frac{2}{1/P+1/R} = \frac{2PR}{P+R}$. A small value has a large reciprocal, pulling the mean down. $P = 1$, $R = .10$: HM $= .20/1.10 \approx .182$. Balanced $P = R = .55$: HM = AM = .55.

$$\text{AM} - \text{HM} = \frac{(P-R)^2}{2(P+R)} \geq 0 \quad \text{(derived step by step in A4)}.$$

HM is a design choice, not a requirement. F1 treats P and R symmetrically; use explicit costs when error consequences are known.

Slide 22 F1 combines precision and recall

$$F_1 = \frac{2\,\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Substitute $\frac{TP}{TP+FP}$ and $\frac{TP}{TP+FN}$ and cancel: numerator $\frac{2TP^2}{(TP+FP)(TP+FN)}$; denominator $\frac{TP[(TP+FN)+(TP+FP)]}{(TP+FP)(TP+FN)}$. Divide:

$$F_1 = \frac{2TP}{2TP+FP+FN}. \qquad \tau = .50: \ \frac{2(3)}{2(3)+2+1} = \frac{6}{9} = \frac{2}{3}.$$

Check: $\frac{2(0.6)(0.75)}{1.35} = \frac{0.9}{1.35} = \frac{2}{3}$ ✓. F1 **excludes TN** and doesn't encode costs.

Slide 23 Rates use different reference groups

$$\text{TPR} = \text{Recall} = \frac{TP}{TP+FN} \ \text{(of all actually impassable roads, what fraction got a warning?)}$$

$$\text{FPR} = \frac{FP}{FP+TN} \ \text{(of all actually passable roads, what fraction got a warning?)}, \qquad \text{Specificity} =$$

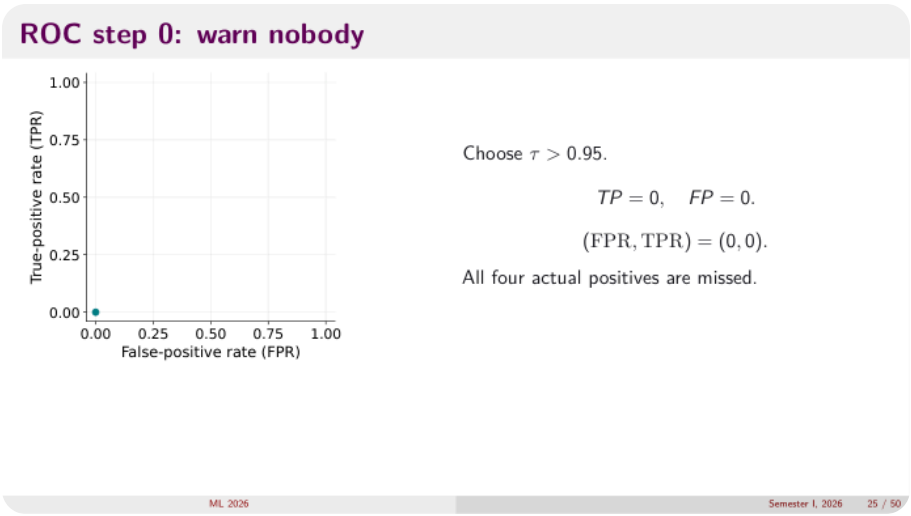
These denominators count actual classes, not predicted classes (so they don't depend on prevalence; precision does).

Slide 24 Each threshold gives one ROC point

τ	TP	FP	FPR	TPR
.50	3	2	2/4 = .50	3/4 = .75
.40	4	2	.50	1.00

Horizontal = FPR, vertical = TPR. An ROC curve collects these as τ varies. **ROC = receiver operating characteristic** (from WWII radar operators tuning a receiver's threshold).

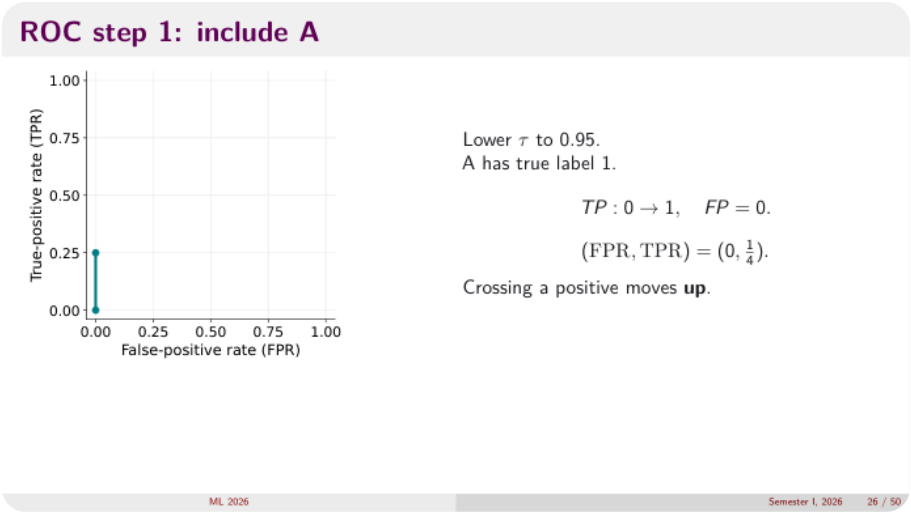
Slide 25 ROC step 0: warn nobody



Slide 25 — τ > 0.95: the point (0, 0).

$\tau > 0.95$: TP = 0, FP = 0 ⇒ (FPR, TPR) = (0, 0). All four positives missed.

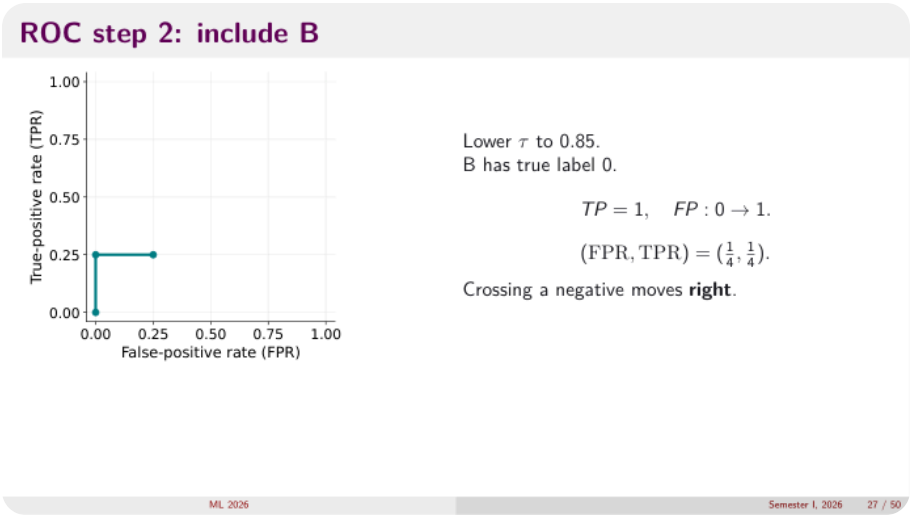
Slide 26 ROC step 1: include A



Slide 26 — A is positive: move up to (0, ¼).

Lower τ to .95. A has label 1: TP 0 → 1, FP = 0 ⇒ $(0, \frac{1}{4})$. **Crossing a positive moves up.**

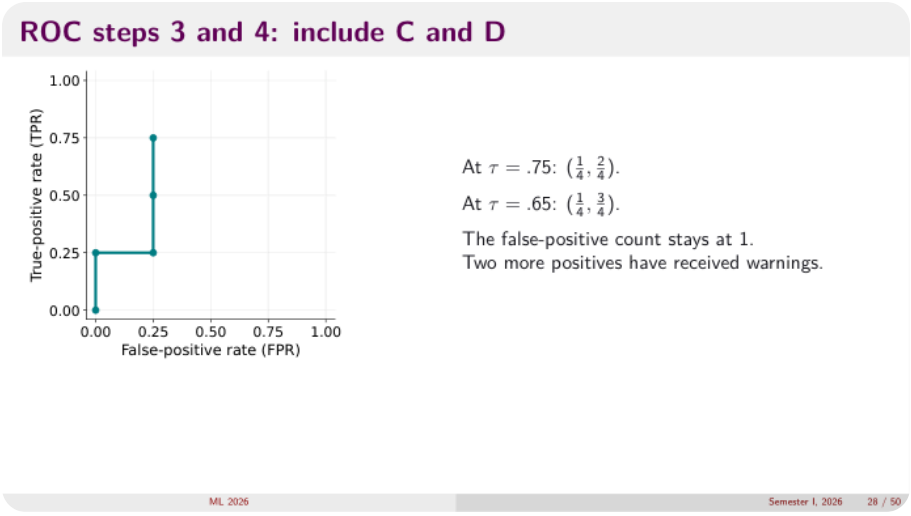
Slide 27 ROC step 2: include B



Slide 27 — B is negative: move right to (¼, ¼).

$\tau = .85$. B has label 0: TP = 1, FP 0 → 1 ⇒ $(\frac{1}{4}, \frac{1}{4})$. **Crossing a negative moves right.**

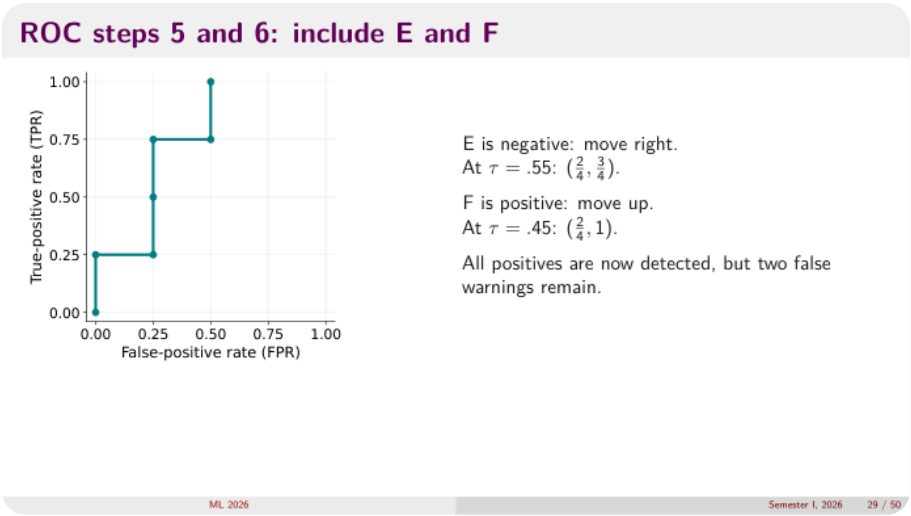
Slide 28 ROC steps 3 and 4: include C and D



Slide 28 — C and D are positives: up to (¼, ½) then (¼, ¾).

$\tau = .75$: $(\frac{1}{4}, \frac{2}{4})$. $\tau = .65$: $(\frac{1}{4}, \frac{3}{4})$. FP stays 1; two more positives warned.

Slide 29 ROC steps 5 and 6: include E and F

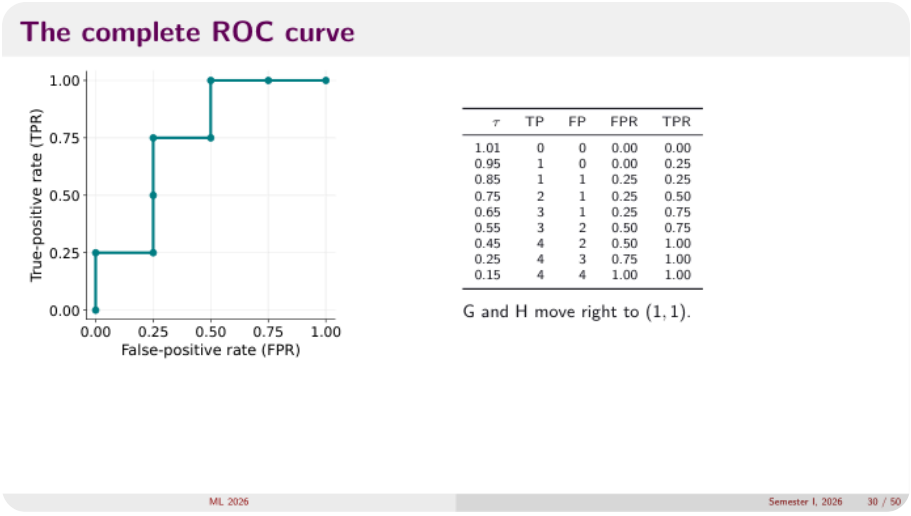


Slide 29 — E (negative) right to (½, ¾); F (positive) up to (½, 1).

E negative → right: $\tau = .55$: $(\frac{2}{4}, \frac{3}{4})$. F positive → up: $\tau = .45$: $(\frac{2}{4}, 1)$. All positives detected, two false warnings remain.

Slide 30

The complete ROC curve

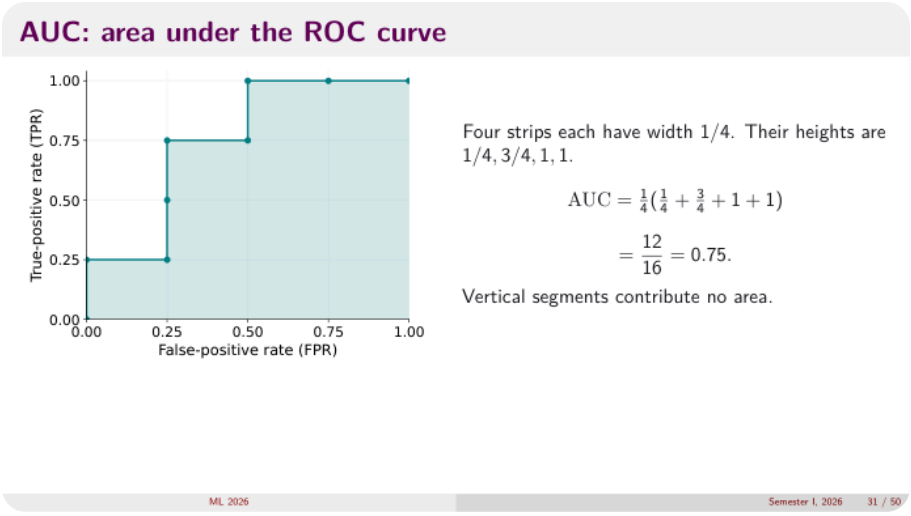


Slide 30 — The complete staircase and its table.

τ	1.01	.95	.85	.75	.65	.55	.45	.25	.15
TP	0	1	1	2	3	3	4	4	4
FP	0	0	1	1	1	2	2	3	4
FPR	0	0	.25	.25	.25	.50	.50	.75	1
TPR	0	.25	.25	.50	.75	.75	1	1	1

G and H move right to (1, 1).

Slide 31 AUC: area under the ROC curve

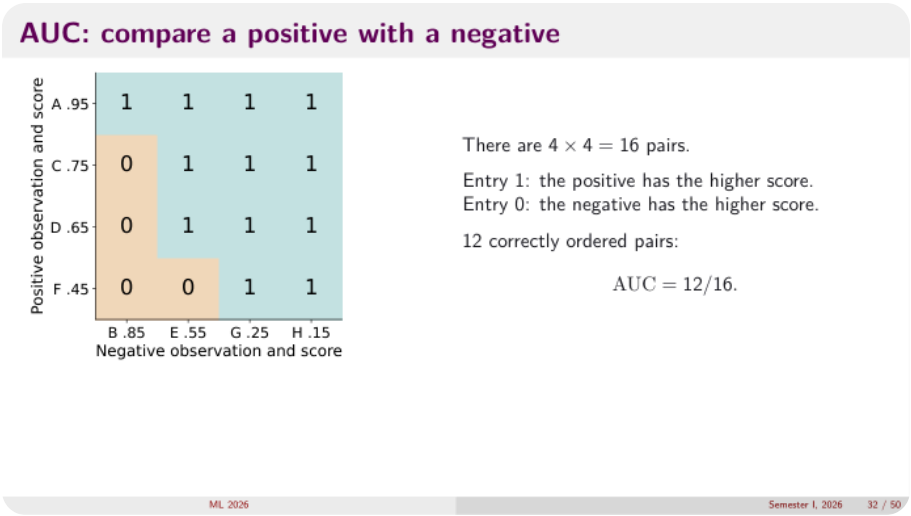


Slide 31 — Four strips of width ¼ with heights ¼, ¾, 1, 1.

$$\mathrm{AUC} = \tfrac{1}{4}\left(\tfrac{1}{4} + \tfrac{3}{4} + 1 + 1\right) = \tfrac{1}{4} \cdot 3 = \tfrac{12}{16} = 0.75.$$

Each strip is under one rightward step (one negative); its height is the TPR reached before that negative. Vertical segments contribute no area.

Slide 32 AUC: compare a positive with a negative



Slide 32 — 4×4 grid of (positive, negative) pairs: 1 if the positive scores higher.

positive \ negative	B .85	E .55	G .25	H .15	wins
A .95	1	1	1	1	4
C .75	0	1	1	1	3
D .65	0	1	1	1	3
F .45	0	0	1	1	2

12 of 16 ⇒ AUC $= 12/16 = 0.75$, the same as the area (L9 A6 explains why). B (.85) beats three positives: it alone costs 3 of the 4 lost pairs.

Slide 33 The general AUC formula

$$\text{AUC} = P(s_+ > s_-) + \tfrac{1}{2}P(s_+ = s_-), \qquad \text{finite sample: } \text{AUC} = \frac{\text{wins} + \frac{1}{2}\text{ties}}{n_+ n_-}.$$

AUC measures ranking. It is not the fraction of correct classifications. (Accuracy at τ = .5 is 5/8 = 0.625; AUC is 0.75; different questions.)

Slide 34 What AUC does not tell us

- **Which threshold?** AUC doesn't select one.
- **How costly are errors?** Not specified.
- **Is a score of .80 really 80%?** Not checked.

If every score becomes s^2, the ranking and AUC stay the same, but the **same numerical threshold** can change decisions: at $\tau = .5$, squared scores $.9025, .7225, .5625, .4225, \ldots$ warn only A, B, C (instead of A–E).

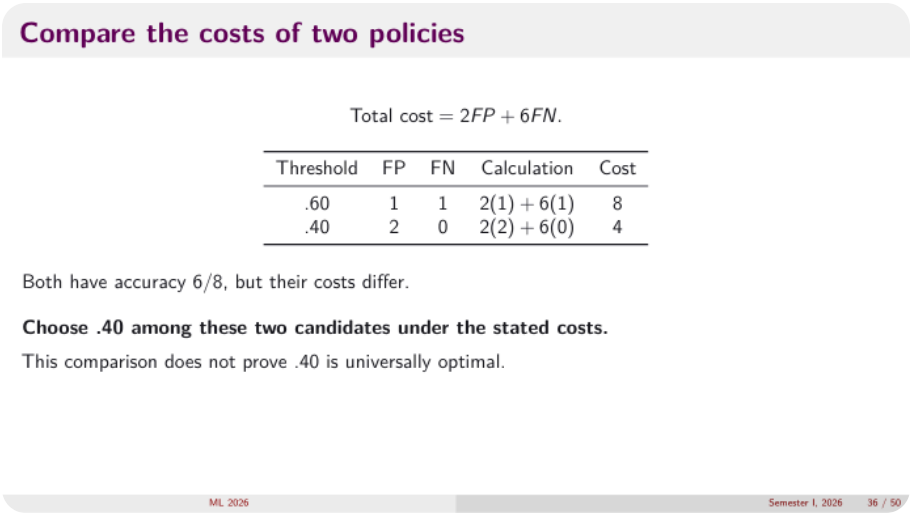
Ranking, probability calibration, and decision quality are different.

Slide 35 State the costs before choosing a threshold

action	actually impassable	actually passable
issue warning	0	$C_{FP} = 2$
do not warn	$C_{FN} = 6$	0

A missed impassable road costs **three times** a false warning. Illustrative assumptions, not measured safety valuations.

Slide 36 Compare the costs of two policies



Slide 36 — Total cost 2FP + 6FN at τ = .60 and .40.

τ	FP	FN	calculation	cost
.60	1 (B)	1 (F)	$2(1) + 6(1)$	8
.40	2 (B, E)	0	$2(2) + 6(0)$	4

Both have accuracy 6/8, but costs differ. Among these two, choose .40. This doesn't prove .40 is universally optimal (the A1 table confirms 4 is the minimum on these 8 roads).

Slide 37 A probability lets us compare actions directly

For a new road with features x, suppose $p = P(Y = 1 \mid x)$ is **reliable for the deployment population**. We don't know the road's true state, so compare the **expected cost** of each action. Correct decisions cost 0, false warning 2, missed event 6.

Slide 38 Expected cost of issuing a warning

If we predict 1: with probability p impassable, cost 0; with probability $1 - p$ passable, cost C_{FP}.

$$R(1 \mid x) = 0 \cdot p + C_{FP}(1 - p) = C_{FP}(1 - p).$$

($R(1 \mid x)$ = expected cost of action 1 for this road.)

Slide 39 Expected cost of not issuing a warning

If we predict 0: with probability p impassable, cost C_{FN}; with probability $1 - p$, cost 0.

$$R(0 \mid x) = C_{FN}\, p.$$

Choose the action with the smaller expected cost.

Slide 40 Derive the probability threshold

Warn when $R(1 \mid x) \leq R(0 \mid x)$:

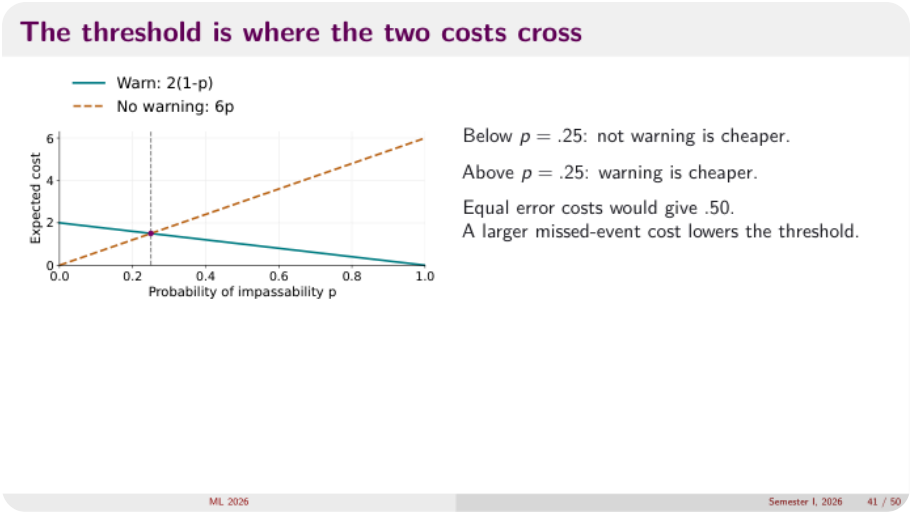
$$C_{FP}(1 - p) \leq C_{FN}p \;\Rightarrow\; C_{FP} \leq (C_{FP} + C_{FN})p \;\Rightarrow\; p \geq \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

$$\tau^* = \frac{2}{2 + 6} = \frac{1}{4}.$$

At equality both actions cost the same; the rule chooses 1.

Check at $p = 0.3$: warn costs $2(0.7) = 1.4$; don't warn costs $6(0.3) = 1.8$ ⇒ warn ✓ (0.3 ≥ 0.25).

Slide 41 The threshold is where the two costs cross



Slide 41 — Warn: 2(1 – p); no warning: 6p; they cross at p = .25.

Below $p = .25$ not warning is cheaper; above it warning is cheaper. Equal costs would give .50. **A larger missed-event cost lowers the threshold.**

Slide 42 Why not just pick the best sample threshold?

The analytic rule .25 minimises **expected** cost under the probability assumptions. On our 8 observations, $\tau = .25$ warns A–G: FP = 3, FN = 0, cost 6; $\tau = .40$ gives cost 4. No contradiction:

- eight realised outcomes are a **noisy sample**, not expected costs;
- the toy scores have **not been shown to be calibrated** probabilities.

Use validation data to choose a practical policy; reserve test data for evaluation.

Slide 43 Population composition changes total cost

$\pi = P(Y = 1)$, population N. Expected negatives $N(1 - \pi)$, expected FP $N(1 - \pi)\mathrm{FPR}$; expected positives $N\pi$, expected FN $N\pi(1 - \mathrm{TPR})$. Assumes the same class-conditional TPR and FPR apply (A7).

Slide 44 Derive cost per observation

$$\text{Expected total cost} = C_{FP} N(1-\pi)\text{FPR} + C_{FN} N\pi(1-\text{TPR}).$$

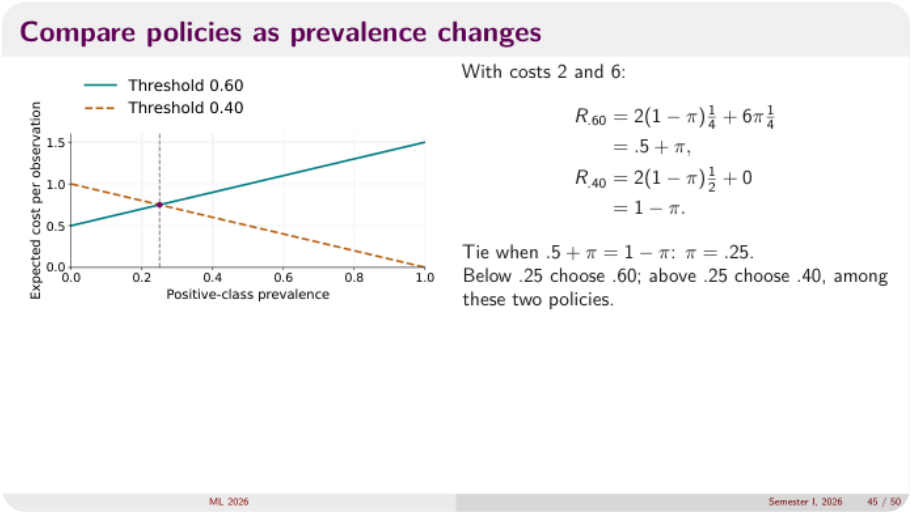
Divide by N:

$$R = C_{FP}(1-\pi)\text{FPR} + C_{FN}\pi(1-\text{TPR}).$$

Averages over both classes; explains why the same TPR/FPR can have different consequences in different populations.

Check on the 8 roads ($\pi = \frac{1}{2}$), τ = .40: $R = 2(\frac{1}{2})(\frac{1}{2}) + 6(\frac{1}{2})(0) = 0.5$; ×8 = 4 ✓.

Slide 45 Compare policies as prevalence changes



Slide 45 — Expected cost per observation vs prevalence for τ = .60 and τ = .40.

τ = .60: FPR = ¼, TPR = ¾. τ = .40: FPR = ½, TPR = 1.

$$R_{.60} = 2(1-\pi)\tfrac{1}{4} + 6\pi\tfrac{1}{4} = 0.5 - 0.5\pi + 1.5\pi = 0.5 + \pi,$$

$$R_{.40} = 2(1-\pi)\tfrac{1}{2} + 6\pi \cdot 0 = 1 - \pi.$$

Tie: $0.5 + \pi = 1 - \pi \Rightarrow \pi = 0.25$. **Below 0.25 choose .60** (fewer false alarms matter more when positives are rare); **above 0.25 choose .40**.

Slide 46 Extension: equal-cost lines on the ROC plot

u = FPR, v = TPR, fixed cost r: $r = C_{FP}(1-\pi)u + C_{FN}\pi(1-v)$. Rearranged (positive costs, $0 < \pi < 1$):

$$v = 1 - \frac{r}{C_{FN}\pi} + \frac{C_{FP}(1-\pi)}{C_{FN}\pi}u.$$

Slope $= C_{FP}(1-\pi)/(C_{FN}\pi)$**.** For a fixed FPR, higher TPR means lower cost. For an empirical ROC, compare achievable points; a smooth tangency isn't required. (8 roads: slope $\frac{2 \cdot 0.5}{6 \cdot 0.5} = \frac{1}{3}$; the best point is $(\frac{1}{2}, 1)$, i.e. τ = .40 — see the Step 3 panel in A7.)

Slide 47 Training weights and decision thresholds differ

Weighted training changes the fitting objective:

$$-\sum_i \left[w_1 y_i \log p_i + w_0(1-y_i)\log(1-p_i)\right], \quad w_1, w_0 > 0.$$

Threshold selection changes the action rule for an already-fitted model. **Class weights don't automatically turn outputs into deployment probabilities.** Don't assume weighting and lowering the threshold are interchangeable in a fitted, regularised model (A8).

Slide 48 A practical evaluation protocol

1. Fit preprocessing and the model on **training** data.
2. Use **validation** data to compare models and choose a threshold; assess calibration if probabilities drive costs.
3. **Freeze** the model, preprocessing, and decision rule.
4. Report **test-set** confusion counts, TPR, FPR, AUC and the stated cost.

Report sample size, class prevalence, and the positive-class definition. An 8-observation teaching example can't establish deployment reliability.

Slide 49 Check the reasoning, not just the formula

1. *Does lowering a threshold retrain the model?* No: the scores are fixed; only the decisions change.
2. *Why does a negative observation move ROC rightward?* Passing it adds one FP, raising FPR by $1/n_-$; TP (hence TPR) is unchanged.
3. *Can two thresholds have equal accuracy but different costs?* Yes: τ = .60 and .40 both have accuracy 6/8, costs 8 vs 4 (slide 36).
4. *Does a high AUC guarantee good probability estimates?* No: AUC depends only on order; any monotone distortion keeps it (slide 34).
5. *If missed positives become more costly, which way should the threshold move?* **Down**: $\tau^* = C_{FP}/(C_{FP} + C_{FN})$ falls as C_{FN} grows.

Slide 50 The formulas and what they answer

formula	answers
$\text{TPR} = \frac{TP}{TP+FN}$, $\text{FPR} = \frac{FP}{FP+TN}$	how each actual class is treated
$\text{AUC} = \frac{\text{wins}+\frac{1}{2}\text{ties}}{n_+ n_-}$	how well the scores rank
observed cost $= C_{FP}FP + C_{FN}FN$	what this sample's decisions cost
$\tau^* = \frac{C_{FP}}{C_{FP}+C_{FN}}$	best threshold for calibrated p, zero cost for correct decisions
$R = C_{FP}(1-\pi)\text{FPR} + C_{FN}\pi(1-\text{TPR})$	expected cost per observation in a population with prevalence π

Count errors; assess ranking; choose actions using consequences.

Part C Practice

1. Using the 8 roads, give the confusion matrix, precision, recall and F1 at $\tau = 0.70$.
2. Swap the labels of B and C (B becomes positive, C negative). Recompute AUC by pair counting.
3. Add a ninth road I: $y = 1$, $s = 0.55$ (tied with E). Compute AUC using the ties formula.
4. TPR = 0.95, FPR = 0.05. At what prevalence does precision drop to 0.5?
5. Show that if $C_{FN} = kC_{FP}$, then $\tau^* = 1/(1+k)$. What is τ^* for $k = 9$?
6. With $C_{FP} = 2$, $C_{FN} = 6$, compute the expected cost per observation of "warn everyone" and "warn no one" as functions of π. Where do they cross?
7. For the 8 roads with $C_{FP} = 1$, $C_{FN} = 1$, which threshold(s) minimise total cost? What metric are you then minimising?
8. Compute AM, HM and AM – HM for $P = 0.8$, $R = 0.4$, and check the formula $(P-R)^2/(2(P+R))$.
9. Two models give recall 0.6 on 50 positives; precisions 0.75 and 0.5. How many false warnings does each issue?
10. Why can the PR baseline never be read as 0.5 in general?
11. A colleague trains with $w_1 = 3$, $w_0 = 1$ and then also applies $\tau^* = 0.25$. What might go wrong?
12. Iso-cost slope for $C_{FP} = 2$, $C_{FN} = 6$ at $\pi = 0.1$? Does the preferred operating point move toward lower or higher FPR compared with $\pi = 0.5$?

1. Warned: A (.95), B (.85), C (.75). TP 2 (A, C), FP 1 (B), FN 2 (D, F), TN 3. Precision $2/3$, recall $2/4$, $F_1 = 4/(4+1+2) = 4/7 \approx 0.571$.
2. New positives A(.95), B(.85), D(.65), F(.45); negatives C(.75), E(.55), G, H. Wins: A 4, B 4, D 3 (loses to C), F 2 (loses to C, E). $13/16 = 0.8125$.
3. Positives A, C, D, F, I(.55); negatives B, E, G, H. Wins: A 4, C 3, D 3, F 2, I: beats G, H (2 wins), ties E (½), loses to B. Total $14 + 0.5 = 14.5$ of $5 \times 4 = 20 \Rightarrow 0.725$.
4. $\frac{0.95\pi}{0.95\pi + 0.05(1-\pi)} = 0.5 \iff 0.95\pi = 0.05(1-\pi) \iff \pi = 0.05/1.0 = 0.05$.
5. $\tau^* = \frac{C_{FP}}{C_{FP}+kC_{FP}} = \frac{1}{1+k}$. $k = 9$: 0.1.
6. Warn everyone: FPR = 1, TPR = 1 $\Rightarrow R = 2(1-\pi)$. Warn no one: FPR = 0, TPR = 0 $\Rightarrow R = 6\pi$. Cross at $2 - 2\pi = 6\pi \Rightarrow \pi = 0.25$ (= τ^*, not a coincidence).
7. Cost = FP + FN = number of errors: minimised where accuracy is max, 6/8, at τ = .65 (or .60) and τ = .45 (or .40). You're minimising the error rate (= 1 – accuracy).
8. AM 0.6; HM $0.64/1.2 = 0.533$; gap 0.0667; formula $0.16/2.4 = 0.0667$ ✓.

9. TP $= 30$. Warnings $= 30/0.75 = 40$ and $30/0.5 = 60$ ⇒ 10 and 30 false warnings.
10. A label-independent rule has precision equal to the prevalence π; it's 0.5 only for balanced classes.
11. Weighting already shifts p upward for positives (the model is no longer calibrated for the deployment population), so applying the cost threshold on top double-counts the cost asymmetry: too many warnings.
12. Slope $\frac{2(0.9)}{6(0.1)} = 3$ (vs $1/3$ at $\pi = 0.5$): much steeper ⇒ the touching point moves toward **lower FPR** (fewer warnings), because false alarms now come from a much larger negative pool.

Previous lectures: L06 (how p is produced) · L09 (the same metrics on 400 roads).